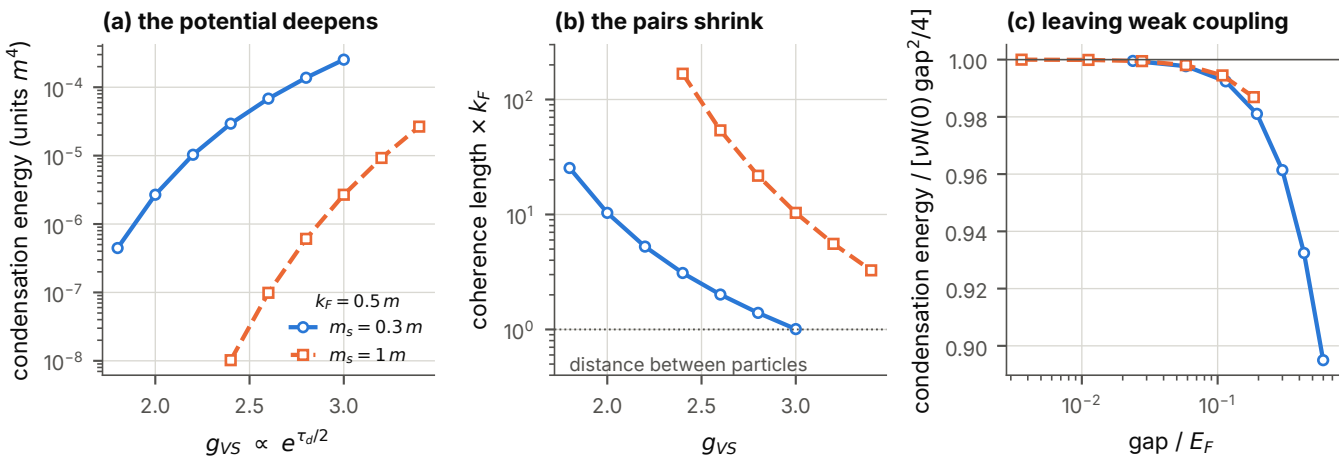
(a) the potential deepens
condensation energy (units m^4)
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
$k_F = 0.5\,m$
$m_s = 0.3\,m$
$m_s = 1\,m$
2.0
2.5
3.0
$g_{VS} \propto e^{\tau_d/2}$
(b) the pairs shrink
coherence length × k_F
10^2
10^1
10^0
distance between particles
2.0
2.5
3.0
g_{VS}
(c) leaving weak coupling
condensation energy / [$\nu N(0)$ gap^2/4]
1.00
0.98
0.96
0.94
0.92
0.90
10^{-2}
10^{-1}
gap / E_F